

High-Speed FBG Interrogation With Electro-Optically Tunable Sagnac Loops

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Abstract—We report high-speed Fiber Bragg Grating (FBG) interrogation based on fiber interferometry. An electro-optical modulator, used as a tunable retarder, is implemented in a fiber Sagnac loop, enabling fast tuning of the interference fringes. Applying an active modulation scheme, the wavelength of the FBG can be retrieved without any requirement of the spectral position of the FBG with respect to the fringes, with no need of tunable sources or spectral analysis. We report experimental results of wavelength measurements with a bandwidth of 100 kHz (only limited by our data acquisition rate) and a dynamic wavelength resolution of $\sigma = 3.7 \text{ fm}/\sqrt{\text{Hz}}$ (corresponding to a strain resolution of $3.1 \text{ n}\epsilon/\sqrt{\text{Hz}}$). We also show that the system can track a continuous tunable laser sweep of 40 nm along the C-band. The technique is polarization-independent and allows wavelength division multiplexing schemes using only one interferometer for all the sensing points. The system is fully based on low-cost off-the-shelf fiber-optical components, and the free-spectral-range of the interferometer can be easily modified by just changing the length of the polarization-maintaining fiber segment.

Index Terms—Fiber Bragg Grating, fiber interferometer, optical fiber sensor, Sagnac interferometer, wavelength meter.

I. INTRODUCTION

FIBER Bragg gratings are very versatile optical fiber sensors which can respond to many physical parameters such as temperature, strain, vibration, pressure, etc. The main advantages of this technology with respect to electronic-based sensors are their immunity to electromagnetic interference, small size, ease of multiplexing along a single fiber, and the fact of being completely passive at the sensing point. However, FBGs need an optoelectronic instrument to extract the reading from the sensing points. This device, commonly referred to as FBG interrogator, is typically based on spectral analysis within a broad optical band in order to be able to simultaneously read many sensors through a wavelength division multiplexing (WDM) scheme. The spectral analysis is typically performed by using either a tunable laser (such as products from Micron Optics or Smart Fibers) or a broad band source and a spectrometer (Ibsen Photonics or Bayspec). It is worth noting that these techniques limit the detection speed to the laser tuning frequency or to the read-out time of the

spectrometer, rarely reaching sampling rates beyond 10 kHz over the full spectral band of operation.

However, several situations can require faster acquisition speeds, in particular for measurement of strain, vibration or pressure fluctuations. In these cases, many high-speed interrogation schemes have been proposed in the past. For instance, a linear filter can be used to extract the wavelength position [1], which has limitations due to the reduced usable spectral window of the interrogator, as well as to the addition of noise related to intensity fluctuations. Other approaches for fast interrogation exploit wavelength-to-time conversion using a chirped source [2, 3], but the measuring range is also very limited with this technique. Optical interferometers can also be used as FBG interrogators [4]. Unbalanced interferometers have a spectral response consisting of interference fringes which allows them to be used as wave-meters. Interferometers can be built using free-space optics, integrated optics or fiber optics. Free-space optics can yield very good performance, but they require a stable optical bench and therefore imply high costs and sensitivity to misalignment. Integrated-optic technologies such as silicon photonics, III-V semiconductors, or doped glass, can be used to fabricate very compact interferometers, and they have also been proposed as FBG interrogators [5,6], in addition to microring-resonator-based tunable filters [7]. However, these devices are typically polarization-dependent, and in addition they do not allow modification by the user of key parameters like the free-spectral range (FSR), as they have to be established in the design before fabrication.

Finally, optical fiber interferometers have the main advantage of being easy to build using off-the-shelf fiber-optic components, and the FSR can be adjusted simply varying the fiber length of the branches [4], [8]. Mach-Zehnder and Michelson interferometers are possible configurations, but in these cases the optical branches are physically separated in different fibers, thus their phase noise is extremely high unless very good isolation mechanisms are implemented. On the other hand, Sagnac loops using polarization-maintaining fibers (PMF) can exploit fiber birefringence to generate the fringes, providing much higher stability due to the fact that the light in the interferometer branches propagate through the same fiber with orthogonal polarization states. Sagnac loops have been proposed in the past as FBG interrogators [9], requiring however the FBG to be aligned to the maximum slope regions in order for the interrogator to properly work.

A rapidly tunable Sagnac loop would enable a much more robust interrogation scheme without requiring the FBG to be

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aligned to certain spectral bands, and being at the same time compatible with WDM multiplexing using a single interferometer. Examples of such active techniques are the Phase Generated Carrier (PGC) [10], pseudo-heterodyne [11], or serrodyne modulation with quadrature sampling [12]. In this paper, we propose and demonstrate for the first time to our knowledge, fiber-based Sagnac loops characterized by fast tunability, and their use as fast FBG interrogators using a PGC phase demodulation scheme.

II. WORKING PRINCIPLE

Sagnac loops are interferometers where the interfering paths travel with orthogonal polarizations in opposite directions. They consist of a closed loop at the output of a 50% fiber coupler containing a segment of polarization-maintaining fiber (PMF) of length L_{PMF} and a polarization controller, whose role is to change the polarization in such a way to maximize the fringe contrast at the output. Once the internal polarization controller is adjusted, the response is robust to any input polarization variation without needing to readjust the internal controller. The output spectrum has a sinusoidal shape due to the interference between propagation along the fast and slow axes of the PMF segment. The free spectral range (FSR) of the fringes is equal to:

$$\Delta\nu_{FSR} = \frac{c}{\Delta n_{PMF} L_{PMF}} \quad (1)$$

where $\Delta\nu_{FSR}$ is the fringe spacing in Hertz, c the speed of light in vacuum, Δn_{PMF} the birefringence of the polarization maintaining fiber, and L_{PMF} the length of the PMF segment. The generated fringe pattern can then be sent to the FBG under test, so that the reflected signal will be modulated by the interferometer.

The main issue is that the responsivity function of the system depends on the relative positions of the fringes and the FBGs. Regions around the maximum slope provide the maximum responsivity, while close to the maxima or minima, the responsivity fades to zero. One way to solve this problem is to use a 3x3 coupler which generates three sinusoidal signals dephased by 120°, similar to the concept proposed in [13]. However, a Sagnac loop with a 3x3 coupler is not polarization independent [14], which prevents its use in this case; in addition, it requires simultaneous acquisition of three channels, which complicates a possible subsequent WDM scheme for multiplexing. Another way to avoid responsivity fading is to continuously tune the fringe pattern back and forth at a faster speed than the FBG shifts. Piezoelectric actuation can tune the filter, but with bandwidth limitations of typically kHz or tens of kHz, which limits the wavelength interrogation speed to a few kHz. On the other hand, modulators based on Pockels effect (such as lithium niobate-based) can generate phase modulation beyond the GHz range. However, in our configuration what we need is a variable retarder, *i.e.*, a rapidly varying birefringence to modulate the retardation to at least a phase π . Fortunately, the electro-optical coefficients of lithium niobate are strongly polarization-dependent, making it possible to generate phase retardations beyond π by applying moderate voltages. In particular, the electro-optical coefficients of lithium niobate are equal to $r_{33} = 32$ pm/V and $r_{31} = 10$ pm/V, [15] which correspond

to the electro-optical coefficients for polarizations aligned (r_{33}) and perpendicular (r_{31}) to the electric field, respectively. When used as a retarder, the response is proportional to the difference between these two values. Therefore, the voltage needed to generate a phase retardation of π between both polarizations can be expressed as:

$$\begin{aligned} V_{\pi}^{(\text{retarder})} &= \frac{r_{33}}{r_{33} - r_{31}} V_{\pi}^{(\text{phase mod})} \\ &\simeq 1.45 V_{\pi}^{(\text{phase mod})} \end{aligned} \quad (2)$$

where $V_{\pi}^{(\text{retarder})}$ is the voltage necessary to generate a phase retardation of π between both polarizations, and $V_{\pi}^{(\text{phase mod})}$ is the voltage which generates an absolute phase variation of π at the polarization aligned with the electric field (associated with coefficient r_{33}), corresponding to the value reported in a typical phase modulator datasheet. For the specific case of *z*-cut lithium niobate, $V_{\pi}^{(\text{retarder})}$ is only 45% higher than $V_{\pi}^{(\text{phase mod})}$. This means that one can shift the interference fringes appreciably by applying reasonable voltage values.

It is worth mentioning that not all lithium niobate phase modulators are suitable for this functionality. For instance, proton-exchanged lithium niobate modulators typically guide only one polarization of light, which excludes them from this application. On the other hand, Ti-diffused modulators can guide both polarizations, but some have an internal polarizer, which would prevent its use as a retarder.

In order to calculate the FSR of the fringe pattern of the interferometer containing the lithium niobate element, one has to consider not only the birefringence of the lithium niobate waveguide itself, but also the birefringence of the access fibers, which are typically polarization maintaining. Actually, birefringence values of lithium niobate and the PMF have opposite signs when the slow axis is aligned to the electric field in the modulator, which is the most typical case, making it easy to compensate one with the other to generate any desired FSR. Mathematically, the resulting FSR of a combination of PMF and lithium niobate will be given by:

$$\Delta\nu_{FSR} = \frac{c}{\Delta n_{LN} L_{LN} - \Delta n_{PMF} L_{PMF}} \quad (3)$$

where Δn_{LN} is the optical birefringence of lithium niobate (equal to 0.0735 at 1550nm [16]), L_{LN} the length of the modulator, Δn_{PMF} the birefringence of the PMF (equal to $3.825 \cdot 10^{-4}$ in our case, a typical panda fiber), and L_{PMF} the total length of the PMF section in the system, including the input and output pigtails of the modulator. The minus sign in the denominator of Eq. 3 applies when the slow axis is aligned to the electric field of the modulator, which is the most typical case. Considering that birefringence of lithium niobate is about 200 times higher than in panda fibers, a few-centimeter modulator length would require just a few-meter PMF total length to cancel the fringes. Therefore, any FSR can be generated by just setting the right PMF length.

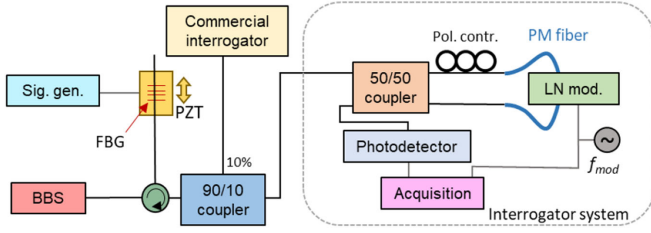


Fig. 1. Experimental setup used for the FBG fast interrogation with active phase demodulation. BBS: Broad-band source; PZT: Piezo-electric transducer; FBG: Fiber Bragg grating; LN: Lithium Niobate. Fiber connections are shown in black while electrical connections are grey.

III. EXPERIMENTAL SETUP

In Fig. 1 we show the experimental setup used for the FBG interrogator measurements with active phase demodulation using a lithium niobate phase modulator. In this particular example, the whole spectrum is sent to the FBG, and the interferometric filter is applied to the signal reflected from the FBG. The reason for this is to have a simultaneous comparison with a commercial interrogator, which required a broad-band source. However, if this comparison were not needed, it would be possible to invert the order and send an already filtered signal to the FBG. This would be convenient in situations where several fiber ports are to be monitored simultaneously, so that one single interferometer can interrogate different fibers with FBGs. In our setup, the broadband source was the spontaneous emission of an Er-doped fiber amplifier, which was attenuated to 15 dBm total power. Light was sent to an FBG clamped to a piezo-electric linear transducer (model PI P-842.60). The FBG was a fs-written-type, with a full-width at half maximum (FWHM) of 600 pm. The reflected beam was redirected through a circulator to a 90/10 coupler, where 10% of the signal was sent to the commercial spectrometry-based interrogator Ibsen photonics model I-MON-512-USB, which has an acquisition sample rate of 3 kHz. Note that only 10% of the reflected signal was sent to the commercial interrogator to avoid its saturation. The other 90% was sent to the Sagnac loop which included a lithium niobate 10 GHz phase modulator with datasheet V_π value of 3.8 V at 50 kHz and insertion loss of 2.9 dB. The modulator was excited with a sinusoidal signal of frequency $f_{mod} = 1.1$ MHz and an amplitude of 10 V peak-to-peak. The modulator had PMF input and output pigtail fibers of a combined total length of ~ 2 m, which was extended with a ~ 5 m PMF patchcord in order to generate a FSR of 2.95 nm. Finally, the signal was collected with an InGaAs photodiode (model PD-LD PDINP0751FAA-0-0-01) which was connected to a trans-impedance variable-gain amplifier (Femto DHPA-100) with a -1 V bias, gain set to 10^5 V/A, bandwidth of 14 MHz, and AC coupling. The output of the amplifier, together with the reference signal sent to the modulator, was collected with a 2-channel 70-MHz data acquisition card (Model PicoScope 2207BMSO) configured to 25 MS/s acquisition rate, and connected to a PC via USB. It is worth mentioning that the most expensive optical parts of the system, which are the broadband source and the LN modulator, are standard off-the-shelf components available in the market for less than \$2000

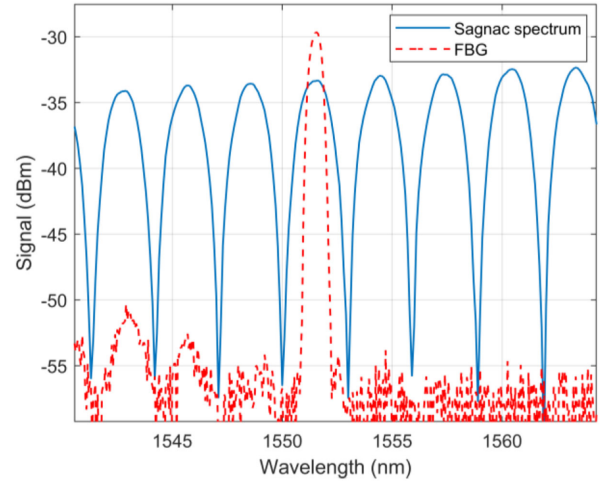


Fig. 2. Blue, solid: Spectrum of the transmission response of the Sagnac loop interferometer with the lithium niobate modulator and a total of ~ 7 m of PMF including the modulator pigtails, so that an FSR of ~ 3 nm was generated. Red, dashed: reflectivity spectrum of the FBG under test, which had a FWHM of ~ 600 pm. Spectra are acquired with an OSA resolution of 70 pm.

USD each, which means that the overall cost of the system can become competitive with most FBG interrogation modules in the market.

Before the experiment, the broad-band source was directly sent to the interferometer and the polarization controller was adjusted to maximize the fringe contrast using an optical spectrum analyzer (if the 50/50 coupler were polarization-maintaining, the polarization controller would have been unnecessary). Fig. 2 shows the Sagnac loop spectrum generated by the lithium niobate modulator together with a total of ~ 7 m PMF including the fiber pigtails. The figure also shows the reflection spectrum of the FBG, with a FWHM of ~ 600 pm. Such bandwidth requires an FSR of the interferometer of at least 1.2 nm to avoid losing contrast in the acquired signal. A FSR of 2.95 nm was generated considering a trade-off between range and responsivity. The interferometer spectrum can be rapidly shifted by the signal sent to the lithium niobate modulator. We experimentally observed that a voltage of 6.5 V shifted the fringes by half an FSR, making our V_π (retarder) equal to 6.5 V.

The extraction of the wavelength with active modulation of the interferometer was performed using a phase generated carrier (PGC) scheme [10]. In this scheme, a sinusoidal input introduced to modulator generates the following response at the output:

$$I(t) = A + B \cos[C \cos(2\pi f_{mod} t) + \psi(t)] \quad (4)$$

where I is the intensity at the detector, A and B are parameters which depend on the signal offset and fringe visibility, C is the semi-amplitude of the modulation depth, f_{mod} the modulation frequency, and ψ is the phase of the signal we are monitoring, which is proportional to the wavelength shift of the FBG. Developing Eq. 4 in its components in f and $2f$ one can univocally extract ψ for example with the arc-tangent method following this equation:

$$\psi = \arctan \left(\frac{J_1(C)I_f}{J_2(C)I_{2f}} \right) \quad (5)$$

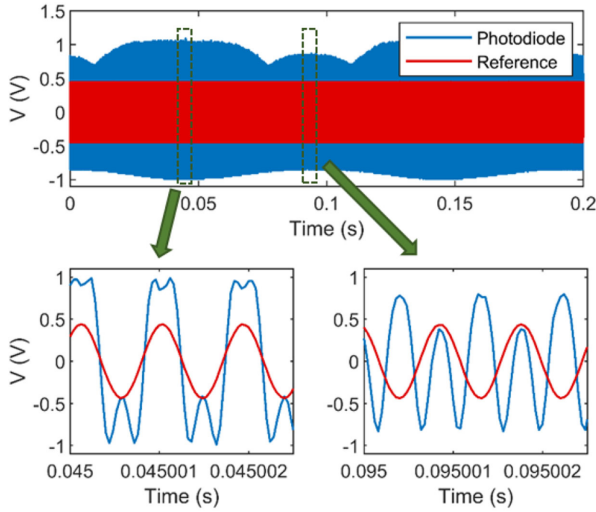


Fig. 3: Raw traces acquired with the acquisition cards. Red: Reference trace generated at $f = 1.1$ MHz, which drives the modulator, divided by 10 to facilitate visualization. Blue: Signal read from the photodiode. Bottom subplots are zoomed views showing different components in f and $2f$, which are used to calculate the phase.

where $J_1(C)$ is the Bessel function of the first-kind, and I_f and I_{2f} are the components in f and $2f$ in the temporal traces, which can be extracted by mixing and low-pass filtering. When $C = 0.84\pi$, $J_1(C)$ becomes equal to $J_2(C)$, which makes them cancel out in Eq. 5.

The processing to extract the wavelength from the temporal traces was the following. A trace of one second duration was acquired at 25 MS/s. The reference signal in f was acquired through the second channel. The signal in $2f$ was digitally generated by squaring the signal in f and subtracting the offset. The acquired photodiode trace was then digitally multiplied by the reference in f and $2f$ and low-pass filtered with a cut-off at 100 kHz to extract I_f and I_{2f} . Finally, Eq. 5 was applied considering a correction factor of $J_1/J_2 = 0.78$, as 10 V peak-to-peak was not sufficient to reach 0.84π of modulation semi-amplitude. Finally, the wavelength was extracted by converting ψ to λ by proportionality considering that $\Delta\psi = 2\pi$ corresponds to a shift of a full FSR. Even though the processing was performed with a PC, a digital electronic system can be engineered to mix the signal with references at f and $2f$ and low-pass filter them in real time. Measurements beyond one FSR can also be implemented by introducing a phase-unwrapping algorithm to the parameter ψ , as shown at the end of the next section.

IV. RESULTS

Fig. 3 shows the raw traces acquired with the acquisition card shown in Fig. 1. The reference is a synthesized sinewave generated at 1.1 MHz, and the photodiode signal shows different components in f and $2f$, which are extracted by mixing to calculate the phase. Fig. 4 shows the result trace applying a 200 Hz pure tone to the FBG. The amplitude was ~ 120 pm peak-to-peak. A very good agreement is observed between the commercial interrogator and the Sagnac-based one. In order to study the possible presence of distortion, a power spectral density analysis

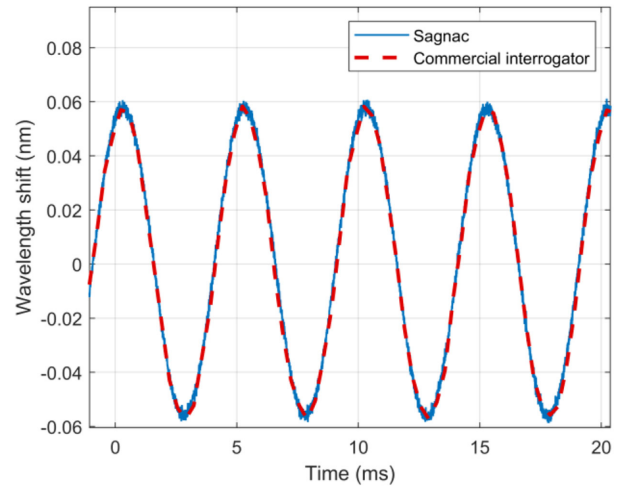


Fig. 4: Temporal trace applying a 200 Hz sinusoidal signal with an amplitude of 120pm peak-to-peak. Sagnac-based result (blue, solid) and commercial interrogator (red, dashed).

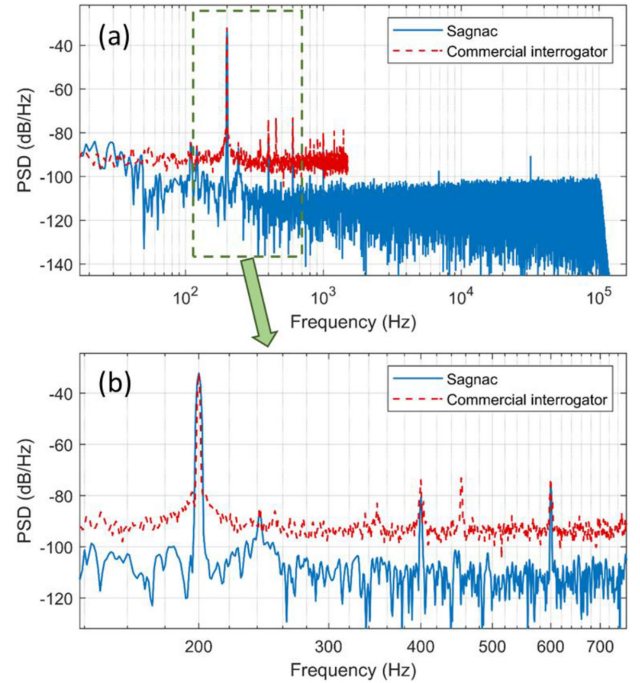


Fig. 5: Power spectral density analysis of the traces of the Sagnac-based interrogator (blue, solid) and the commercial one (red, dashed) applying the 200 Hz sinusoidal signal shown in Fig. 4. (a) Full range; (b) Zoomed range.

of the full trace was performed, and the result is shown in Fig. 5. Fig. 5(a) shows the full trace, where the cut-off frequency of the commercial interrogator at 1.5 kHz is determined by the Nyquist limit imposed by the acquisition rate of 3 kS/s. On the other hand, the frequency cut-off of the Sagnac interferometer is at 100 kHz because that is the cut-off of the low-pass filter applied after the mixing. Fig. 5(b) shows a zoomed section close to the FBG modulation frequency at 200 Hz and its overtones. The weak overtones are due to the nonlinearity of the actuator rather than the demodulation, because they are also observed

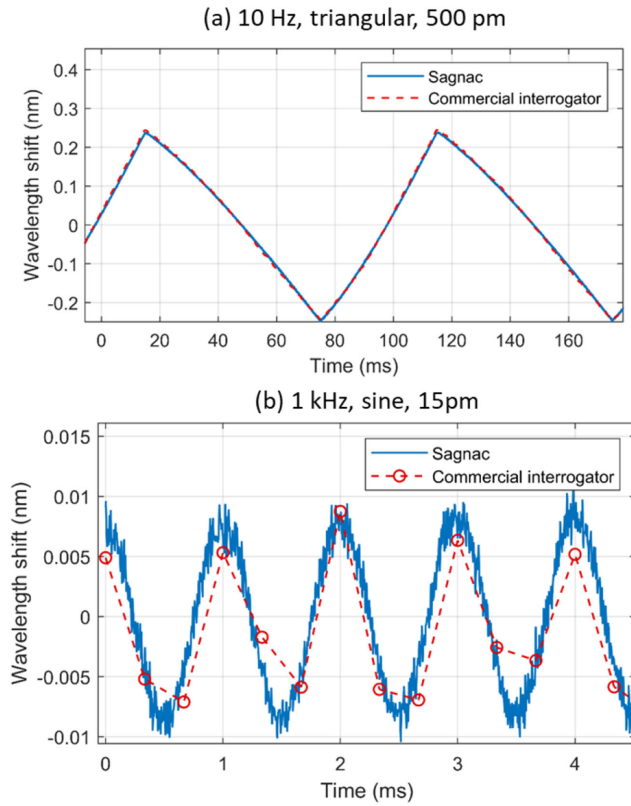


Fig. 6. Different waveforms at different frequencies. (a) a triangular waveform at 10 Hz and 500 pm peak-to-peak amplitude. (b) a sinusoidal waveform at 10 kHz and 15 pm peak-to-peak amplitude. The sampling rate of the commercial interrogator is 3 kS/s.

in the reference measurement. Finally, it is worth noting that the signal-to-noise ratio of the measurement is about 15 dB higher than the reference measurement with the commercial device.

Fig. 6 shows other traces applying different waveforms and frequencies to the FBG. In Fig. 6(a) we show a triangular wave with a frequency of 10 Hz and a large amplitude value of 500 pm peak-to-peak. The wavelength shift is accurately retrieved with the interferometer. On the other hand, in Fig. 6(b) we show a sinusoidal wave of 1 kHz, which is at the frequency limit of our piezo-electric actuator. In this case, the 3 kHz acquisition rate of the commercial interrogator makes it difficult to compare the measured result with the reference, but still, the acquired signal with the Sagnac interferometer shows a clear sine wave.

Fig. 7 shows the noise characterization results. In these measurements, no signal was applied to the FBG. Fig. 7(a) shows the temporal trace of the noise compared to the trace obtained with the commercial interrogator. The $\pm 3\sigma$ margins are also shown for both traces. The standard deviation value for the Sagnac trace is $\sigma = 1.17$ pm, while the commercial interrogator yields $\sigma = 0.95$ pm. Even though the total noise is slightly higher for the interferometer, it is worth mentioning that the signal bandwidths are very different, 100 kHz for the interferometer and 1.5 kHz for the commercial one. In order to calculate the dynamic wavelength resolution, we first did a spectral analysis, which is shown in Fig. 7(b). These traces show a noise behavior

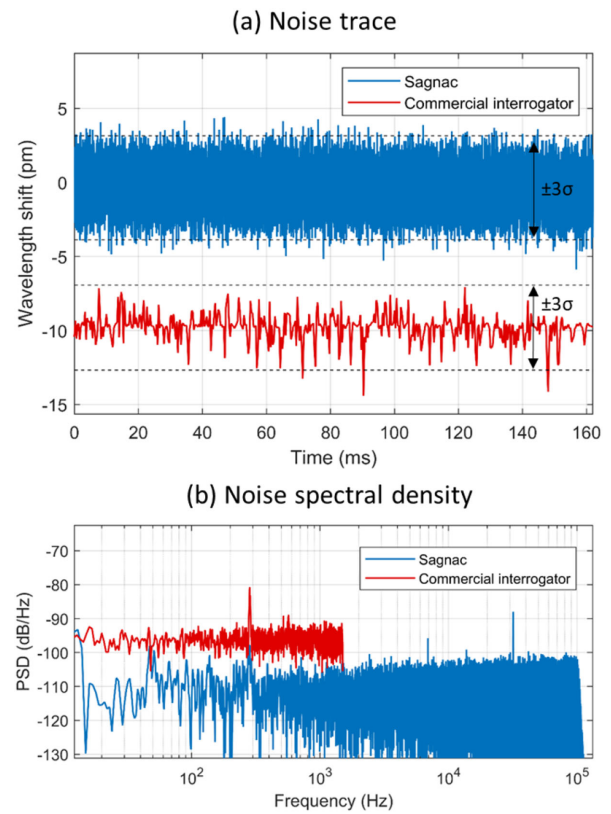


Fig. 7. Noise signals obtained with no actuation on the FBG. (a) Temporal trace of the Sagnac interferometer (blue) and the commercial interrogator (red, with a -10 pm offset for better visualization). (b) Power spectral analysis of the traces, showing a behavior close to white noise.

close to white (flat), which allows us to normalize the dynamic wavelength resolution by dividing by the square root of the bandwidths of the signals, 100 kHz and 1.5 kHz respectively. This calculation yields dynamic wavelength resolution values of 3.7 fm/ $\sqrt{\text{Hz}}$ (Sagnac) and 24 fm/ $\sqrt{\text{Hz}}$ (commercial interrogator) respectively. With a micrometer we also measured the responsivity of the FBG to strain, which was 1.204 pm/ $\mu\epsilon$. This value was used to calculate the dynamic strain resolution, which becomes 3.1 n ϵ / $\sqrt{\text{Hz}}$ (Sagnac) and 19.9 n ϵ / $\sqrt{\text{Hz}}$ (commercial).

All these results show wavelength variations up to 500 pm in amplitude, as the FBG on the piezo-actuator had a limited range. These fluctuations are within a single FSR of the interferometer, which amounted to 2.95 nm in our case. In order to understand whether the system is capable of tracking wavelength variations higher than a FSR, we sent to the interrogator the signal of a tunable laser during a wavelength sweep of 40 nm extension. The laser is an external cavity laser (model EXFO T100S-HP), and was sent directly to the 90/10 coupler shown in Fig. 1. The power of the laser, set to 1 mW, was attenuated down to 300 μW to avoid saturation of the amplifier. The sweep was set between 1530 and 1570 nm at scanning rate of 100 nm/s. Fig. 8 shows the laser sweep acquired with the commercial interrogator and with the Sagnac interferometer. Since the interferometer measures the phase within a fringe, the signal required an unwrapping algorithm which simply consisted in correcting every full-cycle 2π discontinuity by adding 2π offsets. In order to extract the

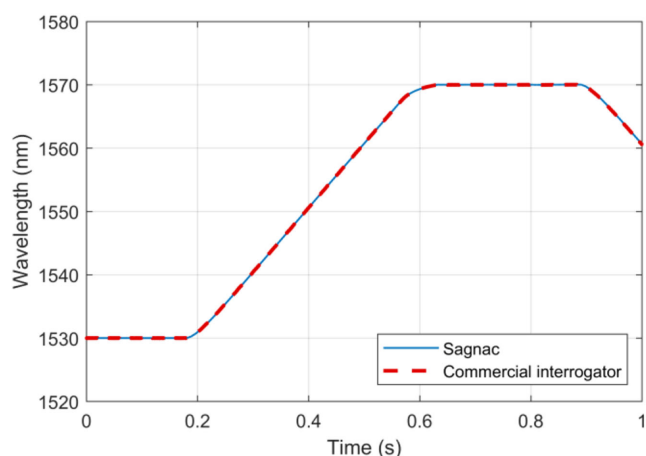


Fig. 8. Acquisition of a 40nm-wide laser sweep between 1530 and 1570 nm (the broad-band source and FBG in Fig. 1 were substituted with a tunable laser). Trace extracted from the interferometer data (blue, solid) and from the commercial interrogator (red, dashed).

absolute wavelength, a final vertical offset was applied to make the trace coincide with the reference trace. Finally, as the range of this measurement is broader, it was necessary to consider a linear variation of the FSR with frequency to account for dispersion. The results show an excellent agreement between both traces, which proves that the wavelength extraction is not limited to a single FSR but can cover several fringes, as long as the number of phase cycles are continuously tracked.

It is worth mentioning that even though Sagnac interferometers are much less sensitive to temperature fluctuations than Mach-Zehnder or Michelson fiber interferometers (because the optical paths are physically common in the former case), the absolute position of the fringes slowly drift when the temperature of the PMF or the modulator varies. This effect is due to birefringence dependence on temperature, and is well described in [9]. This problem can be greatly reduced by using photonic-crystal-type PMFs which have a thermal sensitivity which is ~ 35 times lower than standard Panda-type PMFs [17]. Alternatively, these fluctuations can be canceled by using a reference FBG with respect to which the phase is calculated. The reference FBG can be either on a different fiber, internal to the interrogator, or in the same fiber at a different wavelength and separated by a WDM filter. If the reference FBG is athermal, or its temperature dependence is well characterized and the temperature is measured by other means, the correction would consist in simply subtracting the phase of the FBG under test minus the reference phase, providing a phase value which is independent from thermal fluctuations of the interferometer.

V. CONCLUSION

We proposed and demonstrated a high-speed FBG interrogation scheme based on a rapidly tunable Sagnac interferometer. Fast tunability was achieved by using a commercial dual-polarization lithium niobate phase modulator used as a tunable retarder. Using a PGC-based demodulation scheme, we experimentally demonstrate a detection bandwidth of 100 kHz, only

limited by the acquisition rate of our system. Considering that modulators based on Pockels effect have a bandwidth limitation beyond the GHz, wavelength interrogation can reach sampling rates well beyond hundreds of kHz. The measured dynamic wavelength resolution of the proposed system was $3.7 \text{ fm}/\sqrt{\text{Hz}}$, 6 times better than the commercial system based on spectral peak-detection analysis. Finally, since the PGC demodulation scheme does not require tracking, a single interferometer can monitor many sensors simultaneously by simply adding a WDM filter at the output as in [8].

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